

HMX Grade B



Safety Data Sheet

According To Federal Register / Vol. 77, No. 58 / Monday, March 26, 2012 / Rules And Regulations
Revision Date: 04/07/2015 Date of issue: 04/07/2015

Version: 1.0

SECTION 1: IDENTIFICATION

1.1. Product Identifier

Product Form: Substance

Product Name: HMX Grade B

CAS No: 2691-41-0

Formula: $C_4H_8N_8O_8$

Molecular Weight: 296.2

Holsten SDS Number: 4920

1.2. Intended Use of the Product

Use of the substance/ mixture: Military high explosive. For professional use only.

1.3. Name, Address, and Telephone of the Responsible Party

Company

ORDNANCE SYSTEMS INC [KINGSPORT]

Holston Army Ammunition Plant

4509 West Stone Drive

Kingsport, TN 37660

E-mail contact information: OSIREACHInfo@baesystems.com

1.4. Emergency Telephone Number

Emergency Number : For Emergency Call CHEMTREC® Day or Night
Within USA and Canada: 1-800-424-9300
Outside USA and Canada: +1 703-741-5970 (collect calls accepted)
For more information about this SDS, call OSI Safety (423) 578-6000 or (423) 578-6318

SECTION 2: HAZARDS IDENTIFICATION

2.1. Classification of the Substance or Mixture

Classification (GHS-US)

Expl. 1.1 H201

Acute Tox. 4 (Oral) H302

Acute Tox. 3 (Dermal) H311

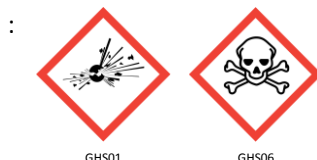
Aquatic Acute 3 H402

Full text of H-phrases: see section 16

2.2. Label Elements

GHS-US Labeling

Hazard Pictograms (GHS-US)



Signal Word (GHS-US)

: Danger

Hazard Statements (GHS-US)

: H201 - Explosive; mass explosion hazard.
H302 - Harmful if swallowed.
H311 - Toxic in contact with skin.
H402 - Harmful to aquatic life.

Precautionary Statements (GHS-US)

: P210 - Keep away from extremely high or low temperatures, ignition sources, and incompatible materials. - No smoking.
P230 - Keep wetted with water.
P240 - Ground/bond container and receiving equipment.
P250 - Do not subject to friction, grinding, shock.
P264 - Wash hands, forearms, and other exposed areas thoroughly after handling.
P270 - Do not eat, drink or smoke when using this product.
P273 - Avoid release to the environment.
P280 - Wear protective gloves, protective clothing, and eye protection.
P301+P330+P312 - IF SWALLOWED: Rinse mouth. Call a poison center or doctor if you feel unwell.
P302+P361+P352+P363- IF ON SKIN: Take off immediately all contaminated clothing. Wash with plenty of water. Wash contaminated clothing before reuse.
P312 - Call a poison center or doctor if you feel unwell.

P370+P380 - In case of fire: Evacuate area.
P372 - Explosion risk in case of fire.
P373 - DO NOT fight fire when fire reaches explosives.
P401 - Store in accordance with local, regional, national, and international regulations.
P405 - Store locked up.
P501 - Dispose of contents/container in accordance with local, regional, national, and international regulations.

2.3. Other Hazards

Exposure may aggravate those with pre-existing eye, skin, or respiratory conditions. Risk of explosion by shock, friction, fire or other sources of ignition. Accidental firing or explosion is likely to cause severe injury or death. Electrostatic charges generated by emptying package in or near flammable vapor may cause flash fire.

2.4. Unknown Acute Toxicity (GHS-US)

No data available

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.1. Substance

Name : HMX Grade B
CAS : 2691-41-0

Name	Product Identifier	%	Classification (GHS-US)
Cyclotetramethylenetetranitramine (HMX)	(CAS No) 2691-41-0	94 - 100	Expl. 1.1, H201 Acute Tox. 4 (Oral), H302 Acute Tox. 3 (Dermal), H311 Aquatic Acute 3, H402

3.2. Mixture

Not applicable

SECTION 4: FIRST AID MEASURES

4.1. Description of First Aid Measures

First-aid Measures General : Never give anything by mouth to an unconscious person. If you feel unwell, seek medical advice (show the label if possible).

First-aid Measures After Inhalation : Remove to fresh air and keep at rest in a position comfortable for breathing. Obtain medical attention if breathing difficulty persists.

First-aid Measures After Skin Contact : Remove contaminated clothing. Drench affected area with water or soap and water for at least 15 minutes. Wash contaminated clothing before reuse. Obtain medical attention if irritation develops or persists.

First-aid Measures After Eye Contact : Rinse cautiously with water for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Obtain medical attention.

First-aid Measures After Ingestion : Do NOT induce vomiting. Rinse mouth. Immediately call a POISON CENTER or doctor/physician.

4.2. Most important symptoms and effects, both acute and delayed

Symptoms/Injuries: Harmful if swallowed or in contact with skin.

Symptoms/Injuries After Inhalation : May cause respiratory irritation. Excessive exposure may cause central nervous system effects may include headache, dizziness, loss of balance and coordination, unconsciousness, coma, respiratory failure, and death.

Symptoms/Injuries After Skin Contact: Harmful in contact with skin. Symptoms may include: Redness, pain, swelling, itching, burning, dryness, and dermatitis.

Symptoms/Injuries After Eye Contact: May cause eye irritation.

Symptoms/Injuries After Ingestion: Harmful if swallowed. Symptoms may include: Weakness. Dizziness. Headache. Nausea. Convulsions. Unconsciousness.

Chronic Symptoms: Chronic exposure to HMX has been reported to cause convulsions or unconsciousness. Chronic local and systemic effects are not fully known. Inhalation and ingestion can result in systemic poisoning, usually affecting the bone marrow (blood-cell-producing system) and the liver. Avoid inhalation and ingestion of dust.

4.3. Indication of Any Immediate Medical Attention and Special Treatment Needed

If you feel unwell, seek medical advice (show the label where possible).

SECTION 5: FIRE-FIGHTING MEASURES

5.1. Extinguishing Media

Suitable Extinguishing Media: DO NOT FIGHT FIRES INVOLVING EXPLOSIVES. Water may be applied through fixed extinguishing system (sprinklers) as long as people need not be present for the system to operate.

Unsuitable Extinguishing Media: DO NOT fight fires involving explosives. Attempts to smother a fire involving this product will be ineffective as it is its own oxygen source. Smothering this product could lead to decomposition and explosion. This product is more sensitive to detonation if contaminated with organic or oxidizable material or if heated while confined. Unless the mass of product on fire is flooded with water, re-ignition is possible.

5.2. Special Hazards Arising From the Substance or Mixture

Fire Hazard: Substance may explode when in contact with flammable or organic substances and confined during a fire. In case of fire involving explosives: Evacuate area. DO NOT fight fires involving explosives. Consult the most current Emergency Response Guidebook (ERG), Guide 112 for additional information. Extreme risk of explosion from shock, friction, fire or other sources of ignition.

Explosion Hazard: Explosives, Division 1.1 - Chemicals and items which have a mass explosion hazard (a mass explosion is one which affects almost the entire quantity present virtually instantaneously).

Reactivity: Reacts violently with many chemicals causing fire and explosion hazard. Material is sensitive to friction, shock, impact, and electrostatic discharge.

5.3. Advice for Firefighters

Precautionary Measures Fire: This product is an explosive with a fire, projection, or blast hazard. DO NOT FIGHT FIRES INVOLVING EXPLOSIVE MATERIALS.

Firefighting Instructions: DO NOT ATTEMPT TO FIGHT FIRE. Immediately evacuate all personnel from the area to a safe distance. Guard against re-entry. Thermal decomposition can lead to release of irritating gases and vapors.

Protection During Firefighting When controlling fire before involvement of explosives, fire-fighters should wear positive pressure self-containing breathing apparatus (SCBA) and full turnout gear.

Other Information: Refer to Section 9 for flammability properties.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal Precautions, Protective Equipment and Emergency Procedures

General Measures: Do not get in eyes, on skin, or on clothing. Do not breathe dust or fumes. Keep away from heat, sparks, open flames, hot surfaces – No smoking. Eliminate every possible source of ignition. Evacuate danger area.

6.1.1. For Non-emergency Personnel

Protective Equipment: Use appropriate personal protection equipment (PPE).

Emergency Procedures: Evacuate danger area.

6.1.2. For Emergency Responders

Protective Equipment: Equip cleanup crew with proper protection.

Emergency Procedures: Upon arrival at the scene, a first responder is expected to recognize the presence of dangerous goods, protect oneself and the public, secure the area, and call for the assistance of trained personnel as soon as conditions permit.

6.2. Environmental Precautions

Prevent entry to sewers and public waters. Hazardous waste due to potential risk of explosion.

6.3. Methods and Material for Containment and Cleaning Up

For Containment: Absorb and/or contain spill with inert material, then place in suitable container. For release to land, contain discharge by constructing dykes or applying inert absorbent; for release to water, utilize damming and/or water diversion to minimize the spread of contamination.

Methods for Cleaning Up: Follow local, state and federal regulations. Clean up spills immediately using a soft bristle brush and a rubber or plastic pan or shovel. Avoid pinching material, metal to metal contact, impact with sharp objects, friction or other situations which may initiate the explosive. Avoid sand, glass, grit, and metal fragments which may sensitize the material to impact and/or friction. Wet with water to desensitize. Use only non-sparking tools. Collect product for recovery or disposal. Collect contaminated soil and water, and absorbent for proper disposal. Notify applicable government authority if release is reportable or could adversely affect the environment.

6.4. Reference to Other Sections

See heading 8, Exposure Controls and Personal Protection. Concerning disposal elimination after cleaning, see item 13.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for Safe Handling

Additional Hazards When Processed: Risk of explosion by shock, friction, fire or other sources of ignition. Must not be confined if burning. Confinement can cause deflagration or transition to detonation with extremely violent results. Explosives may be retained in fissures, cracks, and crevices of structures, equipment and containers which have been exposed to explosives.

Property which may be contaminated by explosives must not be subjected to heat, sparks, or flame. Detonation can occur. Thermal decontamination under controlled conditions is the recommended method for complete decontamination. Thermal decontamination must be preceded by washing/steaming and chemical neutralization or dissolution. Contaminated property must not be buried. During decomposition toxic oxides of nitrogen are emitted.

Precautions for Safe Handling: Explosives must be tested for compatibility with any materials which they contact. Materials include other explosives, solvents, adhesives, metals, plastics, paints, cleaning compound, floor and table coverings, packing materials, and other similar materials and equipment. Keep container closed. Wash thoroughly after handling. Wash contaminated clothing before reuse. Extreme care should be exercised during maintenance of explosive contaminated equipment. Decontamination procedures include washing/steaming, chemical decontamination, and thermal decontamination. Decontamination should be performed prior to welding, cutting or grinding metal parts. Penetrating oil should be used liberally on nuts, bolts, and all threaded connections to aid in desensitizing hidden explosives prior to disassembly.

Hygiene Measures: This product is an explosive and should only be used under the supervision of trained and licensed personnel. Handle in accordance with good industrial hygiene and safety procedures. Wash hands and other exposed areas with mild soap and water before eating, drinking, or smoking and again when leaving work.

7.2. Conditions for Safe Storage, Including Any Incompatibilities

Technical Measures: Store as defined in the provisions of the Bureau of Alcohol, Tobacco and Firearms regulations contained in 27 CFR Part 555.

Storage Conditions: High explosives should be stored in approved explosives magazines in accordance with regulatory requirements. Storage and handling must be carried out in accordance with appropriate Safety Agency regulations concerning quantity distance, barricading, personnel exposure and material handling equipment. Store under moderate temperatures recommended by a technical services representative. Keep away from heat, spark and flames. Keep containers closed. Explosives should be kept well away from initiating explosives; protected from physical damage; separated from oxidizing materials, combustibles, and sources of heat. Isolate from incompatibles. In buildings and locations where explosives with spark energies for initiation not greater than 0.02 Joules are handled, the relative humidity should be 50% or greater. Dust generated by handling must be cleaned up on a continuing basis.

Incompatible Products: Heat sources. Direct sunlight. Acids. Oxidizers. Alkalis. Avoid physical sensitizers such as glass, sand, and metal fragments.

Special Rules on Packaging: Keep only in the original container.

7.3. Specific End Use(s)

Military high explosive. For professional use only.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control Parameters

For substances listed in section 3 that are not listed here, there are no established exposure limits from the manufacturer, supplier, importer, or the appropriate advisory agency including: ACGIH (TLV), NIOSH (REL), or OSHA (PEL).

8.2. Exposure Controls

Appropriate Engineering Controls

: Proper grounding procedures to avoid static electricity should be followed. Emergency eye wash fountains and safety showers should be available in the immediate vicinity of any potential exposure. Product to be handled in a closed system and under strictly controlled conditions. Ensure all national/local regulations are observed.

Personal Protective Equipment

: Gloves. Protective clothing. Safety glasses. Insufficient ventilation: wear respiratory protection.



Materials for Protective Clothing

: For explosive-handling workers, caps and coveralls for full body (arms and legs) protection are recommended. Cotton coveralls, underwear, socks and conductive shoes are recommended to avoid human static discharge.

Hand Protection

: Wear chemically resistant protective gloves.

Eye Protection

: Chemical goggles or safety glasses.

Skin and Body Protection

: Wear suitable protective clothing. Wear long sleeves.

Respiratory Protection

: If exposure limits are exceeded or irritation is experienced, approved respiratory protection should be worn.

Environmental Exposure Controls

: Do not allow the product to be released into the environment.

Consumer Exposure Controls

: Do not eat, drink or smoke during use.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on Basic Physical and Chemical Properties

Physical State

: Solid

Appearance

: White, crystalline

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Odor	: Odorless
Odor Threshold	: No data available
pH	: No data available
Evaporation Rate	: No data available
Melting Point	: 280 °C w/ decomposition (536 °F)
Freezing Point	: No data available
Boiling Point	: No data available
Flash Point	: No data available
Auto-ignition Temperature	: No data available
Decomposition Temperature	: No data available
Flammability (solid, gas)	: No data available
Vapor Pressure	: No data available
Relative Vapor Density at 20 °C	: No data available
Relative Density	: No data available
Specific Gravity	: 1.9 (water = 1)
Solubility	: Insoluble in water
Partition Coefficient: N-Octanol/Water	: No data available
Viscosity	: No data available
Explosive Properties	: Class 1.1 - Explosives (with a mass explosion hazard) 49 CFR 173.21.

9.2. Other Information No additional information available

SECTION 10: STABILITY AND REACTIVITY

- 10.1. Reactivity: Reacts violently with many chemicals causing fire and explosion hazard. Material is sensitive to friction, shock, impact, and electrostatic discharge.
- 10.2. Chemical Stability: HMX is a military high explosive. HMX has been assigned the United Nations Organization Classification of Class 1, Division 1 (mass detonating) based on the Department of Defense Explosives Hazard Classification Procedures, Army Technical Bulletin 700-2.
- 10.3. Possibility of Hazardous Reactions: Hazardous polymerization will not occur.
- 10.4. Conditions to Avoid: Avoid shock, heat, electrostatic discharge, impact, impingement, and friction. High explosive will detonate when exposed to sufficient energy level.
- 10.5. Incompatible Materials: Heat sources. Direct sunlight. Acids. Oxidizers. Alkalis. Avoid physical sensitizers such as glass, sand, and metal fragments.
- 10.6. Hazardous Decomposition Products Thermal decomposition generates: Carbon oxides (CO, CO₂). Nitrogen oxides.

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information On Toxicological Effects

Acute Toxicity: Oral: Harmful if swallowed. Dermal: Toxic in contact with skin.

HMX Grade B	
ATE (Oral)	1,670.00 mg/kg body weight
ATE (Dermal)	982.00 mg/kg body weight
Cyclotetramethylenetetranitramine (2691 -41-0)	
LD50 Oral Rat	1670 mg/kg
LD50 Dermal Rat	982 mg/kg (Species: New Zealand White)

Skin Corrosion/Irritation: Not classified

Serious Eye Damage/Irritation: Not classified

Respiratory or Skin Sensitization: Not classified

Germ Cell Mutagenicity: Not classified

Carcinogenicity: Not classified

Reproductive Toxicity: Not classified

Specific Target Organ Toxicity (Single Exposure): Not classified

Specific Target Organ Toxicity (Repeated Exposure): Not classified

Aspiration Hazard: Not classified

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Symptoms/Injuries After Inhalation : May cause respiratory irritation. Excessive exposure may cause central nervous system effects may include headache, dizziness, loss of balance and coordination, unconsciousness, coma, respiratory failure, and death
Symptoms/Injuries After Skin Contact: Harmful in contact with skin. Symptoms may include: Redness, pain, swelling, itching, burning, dryness, and dermatitis

Symptoms/Injuries After Eye Contact: May cause eye irritation

Symptoms/Injuries After Ingestion: Harmful if swallowed. Symptoms may include: Weakness. Dizziness. Headache. Nausea. Convulsions. Unconsciousness

Chronic Symptoms Chronic exposure to HMX has been reported to cause convulsions or unconsciousness. Chronic local and systemic effects are not fully known. Inhalation and ingestion can result in systemic poisoning, usually affecting the bone marrow (blood-cell-producing system) and the liver. Avoid inhalation and ingestion of dust

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity

Ecology - General : This material is hazardous to the aquatic environment. Keep out of sewers and waterways.

Ecology - Water : Harmful to aquatic life.

Cyclotetramethylenetetranitramine (2691 -41-0)	
LC50 Fish 1	8.8 - 26 mg/l (Exposure time: 96 h - Species: Pimephales promelas [static])
LC 50 Fish 2	> 32 mg/l (Exposure time: 96 h - Species: Lepomis macrochirus [static])

12.2. Persistence and Degradability No additional information available

12.3. Bioaccumulative Potential No additional information available

12.4. Mobility in Soil No additional information available

12.5. Other Adverse Effects

Other Information : Avoid release to the environment.

SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste Treatment Methods: Hazardous waste due to potential risk of explosion. Dispose of waste material in accordance with all local, regional, national, and international regulations.

Waste Disposal Recommendations: Disposal should comply with all federal, state and local regulations. Waste is considered hazardous waste for reactivity characteristic (United States EPA D003 hazardous waste classification). Explosives should be destroyed by open burning, by burning in an approved incinerator, or by chemical treatment with caustics. The disposal site should be located to provide adequate quantity-distance protection for adjacent facilities and personnel. Explosives should not be burned in containers. The explosives should be ignited remotely.

SECTION 14: TRANSPORT INFORMATION

* HMX is FORBIDDEN for transport unless wetted with not less than 15 percent water, by mass.

14.1. In Accordance with DOT

Proper Shipping Name : HMX, WETTED or Octogen, wetted with not less than 15 percent water, by mass

Hazard Class : 1.1D

Identification Number : UN0226

Label Codes : 1.1D

Packing Group : II

Marine Pollutant : Marine pollutant

ERG Number : 112

14.2. In Accordance with IMDG

Proper Shipping Name : CYCLOTETRAMETHYLENETETRANITRAMINE (HMX; OCTOGEN), WETTED

Hazard Class : 1

Identification Number : UN0226

Label Codes : 1.1D

EmS-No. (Fire) : F-B

EmS-No. (Spillage) : S-Y

Marine Pollutant : Marine pollutant



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14.3. In Accordance with IATA

Proper Shipping Name : HMX, WETTED
Identification Number : UN0226
Hazard Class : 1
Label Codes : 1.1D
ERG Code (IATA) : 1L



SECTION 15: REGULATORY INFORMATION

15.1 US Federal Regulations

HMX Grade B	
SARA Section 311/312 Hazard Classes	Immediate (acute) health hazard Sudden release of pressure hazard
Cyclotetramethylenetetranitramine (2691 -41-0)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory	

15.2 US State Regulations

Cyclotetramethylenetetranitramine (2691 -41-0)	
U.S. - New Jersey - Right to Know Hazardous Substance List	

SECTION 16: OTHER INFORMATION, INCLUDING DATE OF PREPARATION OR LAST REVISION

Revision Date : 04/07/2015
Other Information : This document has been prepared in accordance with the SDS requirements of the OSHA Hazard Communication Standard 29 CFR 1910.1200.

GHS Full Text Phrases:

Acute Tox. 3 (Dermal)	Acute toxicity (dermal) Category 3
Acute Tox. 4 (Oral)	Acute toxicity (oral) Category 4
Aquatic Acute 3	Hazardous to the aquatic environment - Acute Hazard Category 3
Expl. 1.1	Explosive Category 1.1
H201	Explosive; mass explosion hazard
H302	Harmful if swallowed
H311	Toxic in contact with skin
H402	Harmful to aquatic life

This information is based on our current knowledge and is intended to describe the product for the purposes of health, safety and environmental requirements only. It should not therefore be construed as guaranteeing any specific property of the product.

SDS US (GHS HazCom)